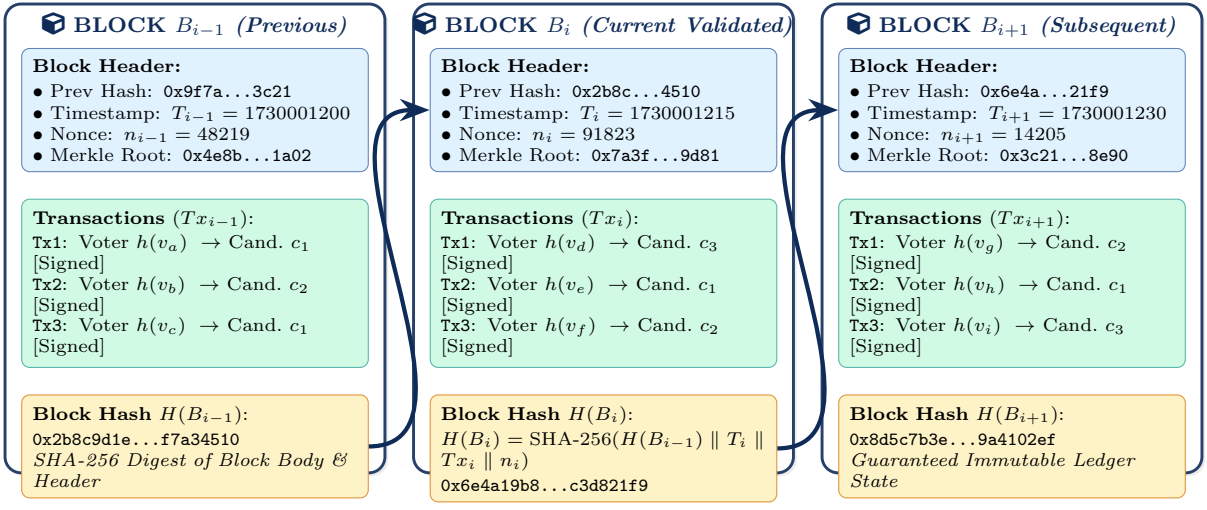

BLOCK B_{i-1} (Previous)
Block Header:
• Prev Hash: 0x9f7a...3c21
• Timestamp: $T_{i-1} = 1730001200$
• Nonce: $n_{i-1} = 48219$
• Merkle Root: 0x4e8b...1a02
Transactions (Tx_{i-1}):
Tx1: Voter $h(v_a)$ → Cand. c_1 [Signed]
Tx2: Voter $h(v_b)$ → Cand. c_2 [Signed]
Tx3: Voter $h(v_c)$ → Cand. c_1 [Signed]
Block Hash $H(B_{i-1})$:
0x2b8c9d1e...f7a34510
SHA-256 Digest of Block Body & Header
BLOCK B_i (Current Validated)
Block Header:
• Prev Hash: 0x2b8c...4510
• Timestamp: $T_i = 1730001215$
• Nonce: $n_i = 91823$
• Merkle Root: 0x7a3f...9d81
Transactions (Tx_i):
Tx1: Voter $h(v_d)$ → Cand. c_3 [Signed]
Tx2: Voter $h(v_e)$ → Cand. c_1 [Signed]
Tx3: Voter $h(v_f)$ → Cand. c_2 [Signed]
Block Hash $H(B_i)$:
$H(B_i) = \text{SHA-256}(H(B_{i-1}) \parallel T_i \parallel Tx_i \parallel n_i)$
0x6e4a19b8...c3d821f9
BLOCK B_{i+1} (Subsequent)
Block Header:
• Prev Hash: 0x6e4a...21f9
• Timestamp: $T_{i+1} = 1730001230$
• Nonce: $n_{i+1} = 14205$
• Merkle Root: 0x3c21...8e90
Transactions (Tx_{i+1}):
Tx1: Voter $h(v_g)$ → Cand. c_2 [Signed]
Tx2: Voter $h(v_h)$ → Cand. c_1 [Signed]
Tx3: Voter $h(v_i)$ → Cand. c_3 [Signed]
Block Hash $H(B_{i+1})$:
0x8d5c7b3e...9a4102ef
Guaranteed Immutable Ledger State